

Karnati Giridhar Reddy

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RESEARCH INTERESTS

RF and microwave instrumentation; wide-band antenna design and electromagnetic simulation; radio receiver characterisation; field-deployable ground-station systems for Satellite telemetry reception.

EDUCATION

Chaitanya Bharathi Institute of Technology (CBIT) Sept 2022 – Jul 2026 (expected)

B.E. in Electronics & Communication Engineering; Minor in Computer Science Hyderabad, India

CGPA: **9.23 / 10.0**; **Rank-1, ECE Department**; Silver Medal for Academic Excellence (AY 2023–24).

Relevant coursework: Antennas & Wave Propagation, Microwave Engineering, RF & Microwave Circuits, Electromagnetic Theory, Communication Systems, Digital Signal Processing, Digital System Design (Verilog).

Sri Chaitanya Junior Kalasala

2020 – 2022

Intermediate (Class XI–XII), Mathematics–Physics–Chemistry

Hyderabad, India

Aggregate: 96.2%.

RESEARCH EXPERIENCE

Raman Research Institute (RRI)

Dec 2025 – April 2026

Visiting Research Student,

Bengaluru, India

- Architectural design and bring-up of a Water-Vapour Radiometer; performed **RF link-budget and thermal-noise analysis**, and gain/noise-figure cascade modelling for the IF chain.

RaoS Consultants LLC

Mar 2025 – Present

Internship Trainee, Antenna Design & RF Hardware

Part-time

Indian Institute of Technology Hyderabad

Sept 2025

Research Intern, Nanofabrication Lab (SPARC NANO-X)

Hyderabad, India

- Trained on the IC design flow from layout through packaging; fabricated nano-silicon waveguides, microfluidic devices, and 50 μm Au inter-digitated electrodes for 2-D-material gas sensors using Heidelberg μPG 501 maskless lithography, DC magnetron sputtering and lift-off in a Class-1000 cleanroom.

Indian Institute of Technology Kharagpur

Jul – Aug 2025

IEEE Summer Internship – RF Phase-Shifter Design

Kharagpur, India

- Designed and simulated microwave phase shifters in Keysight ADS with MATLAB post-processing; evaluated insertion loss, phase error and return loss across band.

Indian Institute of Science (IISc), Bangalore

Jul 2025

Electrical Engineering Summer School (EESS-25)

Bengaluru, India

- Selected as **1 of 100 students nationwide** for an intensive programme in signal processing, communications and microelectronics.

National Institute of Technology Karnataka (NITK), Surathkal*Summer Research Intern, SDR & Embedded Radio Systems*

May – Jun 2025

Surathkal, India

- Built SDR-based transceivers and Raspberry-Pi-driven portable APRS ground nodes for weather monitoring deployable to remote field sites.

CREST, BITS Pilani Hyderabad Campus*Research Trainee, ICAN 4.0 Workshop*

Oct 2024

Hyderabad, India

- Fabricated and electrically characterised 2-D-material-based nano-device sensors.

SELECTED RESEARCH & ENGINEERING PROJECTS**Wide-Band Blade Dipole Antenna for UAV applications***Ansys HFSS, CST Studio Suite*

2025

Independent

- Designed and electromagnetically simulated a wide-band blade (bow-tie) dipole on a finite ground plane, targeting the UHF / sub-GHz band relevant to UAV, bird like drone applications.
- Achieved $|S_{11}| \leq -10$ dB across approximately **0.55–1.51 GHz** (bandwidth ≈ 960 MHz; fractional bandwidth $\approx 93\%$); deep resonance at 0.73 GHz with $|S_{11}| = -39.73$ dB, corresponding to **VSWR $\approx 1.02:1$** . VSWR $< 2:1$ was maintained across the entire operational band.
- Cross-validated in CST Studio (Frequency-Domain solver), confirming consistent EM behaviour across the two solvers – a low-loss, smooth-impedance radiator suitable for spectrally-clean 21 cm observations.

CubeSat Antenna Deployment Sub-System – Burn-Wire Mechanism*Industrial Project, CBIT*

2024 – 2025

Supervisor: Dr. K. Vasanth

- Designed and prototyped a compact, low-power burn-wire deployment sub-system for a 1U CubeSat with a four-stage release sequence (initial restriction, motion release, deployment guide, final latching).
- Built a custom relay-controlled over-current PCB driving a resistive burn element through a nylon restraint, with a temperature-sensor safety interlock and microcontroller arming logic; selected shape-memory-alloy radiating arms for repeatable self-erection.

LoRa-Based Satellite Ground Station – VayuVani*Antenna Design, RF Hardware, Python Telemetry Pipeline*

2024 – 2025

IEEE Xplore

- Designed, fabricated and field-tested a 14-wave Ground-Plane antenna and supporting Dipole, Yagi–Uda, Quadrifilar-Helical and Turnstile antennas for a 433 MHz LoRa CubeSat downlink; performed comparative link-margin analysis.
- Built the **VayuVani** ground station with custom Python decoders for LoRa telemetry and offline data archival – received the **IEEE Best Paper Award** at SCEEECS 2025.

SatNOGS Ground Station – Distributed Satellite Data Acquisition*Open-Source Network Contribution*

Ongoing

Hyderabad

- Operates an automated SatNOGS station contributing to the global open-source satellite network; managed scheduled passes for CubeSats, AMSATs and the ISS (ARISS), including time-critical end-of-life events.
- Field experience with site-survey, antenna mounting, RF cabling/grounding and noise-floor characterisation in non-laboratory environments

Multi-Core Soft-Processor Cluster*Verilog HDL, Xilinx Vivado*

2025

- Implemented a multi-core soft-processor cluster in Verilog with LFSR-based traffic generators and a round-robin arbiter FSM, targeted for FPGA synthesis. Additional digital-design portfolio: 32-bit and carry-skip adders, barrel shifter, UART, BCD-to-7-segment, clock divider.

Other Projects

- **VHF Wireless Beacon** for high-altitude balloon and CanSat missions (Arduino Nano + DRA818V); adapted for amateur-radio direction-finding.
- **ADS-B Receiver System** contributing live aircraft data to global aggregator networks.
- **CMOS 180 nm process-to-device simulation** in Silvaco TCAD; extracted V_t , I_{dsat} , I_{off} , SS, DIBL.
- **Schottky-diode forward I–V** simulation with thermionic-emission, SRH, Auger and bandgap-narrowing models.

PUBLICATIONS

1. **K. Giridhar Reddy**, “VayuVani: A Self-Managed LoRa Ground Station for Amateur Operators,” *HAM Fest India 2025 Magazine*, Goa, December 2025, pp. 20–27.
hf2025.in/hfimag.pdf
2. **K. Giridhar Reddy et al.**, “Design of Ground Station Antennas for Tracking of LoRa-Based Satellites at 433 MHz Amateur Frequencies,” in *Proc. IEEE International Students’ Conference on Electrical, Electronics and Computer Sciences (SCEEECS 2025)*, NIT Bhopal, February 2025; published on IEEE Xplore, 2 April 2025. **Best Paper Award**.
ieeexplore.ieee.org/document/10940725
3. **K. Giridhar Reddy**, “Privacy-Preserving LoRa Ground Stations in Amateur Satellite Communications: A Case Study of VayuVani,” *Radio Amateurs’ World (RAW) Magazine*, presented at HAM Fest India 2024, Kolkata, 21 December 2024.

TECHNICAL SKILLS

EM & RF Simulation	Ansyz HFSS, CST Studio Suite, Altair FEKO, Keysight ADS, MATLAB RF Toolbox
RF & Microwave	Wide-band antennas, LNA, mixer, PLL/VCO, power amplifier, filter design, link-budget analysis, noise-figure / IIP3, anechoic-chamber and VNA measurement workflows
RF Hardware & Bench	VNA, spectrum analyser, SDR (RTL-SDR, HackRF), soldering and RF connectorisation, PCB design and prototyping; licensed amateur-radio operator (VU2TIO)
Digital, VLSI & EDA	Cadence Virtuoso, Xilinx Vivado, Silvaco TCAD (Victory Process / Device / Extract), CleWin, SolidWorks
Programming & HDL	Python, C/C++, MATLAB, Verilog, VHDL, LabVIEW
Embedded	Arduino, Raspberry Pi, ESP32 (Xtensa LX6), RISC-V, Arm Cortex-M; FPGA prototyping
Fabrication	Photolithography, maskless laser writing, DC magnetron sputtering, PVD, CVD, lift-off, RTA; I–V (Keithley 2450)

HONOURS & AWARDS

- GATE 2025 & 2026 qualified, Electronics & Communication Engineering (Score 436, GATE 2025).
- Best Paper Award, IEEE SCEEECS 2025, NIT Bhopal.
- Rank-I, ECE Department & Silver Medal for Academic Excellence, CBIT (AY 2023–24).
- Selected for IISc Electrical Engineering Summer School (EESS-25) – top 100 nationwide.
- 2nd Prize, Technical Quiz, NAFED-o8 Conference (VSSC / ISRO, hosted at IIT Hyderabad).
- Three-time NPTEL Star Winner (Enthusiast, Discipline Star, Believer), IIT Madras.

POSITIONS OF RESPONSIBILITY

- **Founder & President**, HAM Radio Club, CBIT (Sept 2023 – Present) – led technical projects and STEM-outreach workshops for school students.
- **Technical Chair & Coordinator**, SYNAPSE 2025, CBIT – managed all technical events for the college techno-cultural festival.

REFERENCES

Available upon request. Recommenders include faculty and scientists at CBIT Hyderabad, NIT Surathkal, Rao's Consultants LLC, and IIT Hyderabad's Nanofabrication Lab.